

AVERAGE SALARY BY DEPARTMENT

Problem Statement

TechNova Solutions wants to analyze salary distribution across various departments.

The HR team needs a report showing the average salary of employees in each department to help with compensation planning and budgeting.

Task

Write an SQL query to calculate the average salary of employees in each department.

The output must contain:

- department
- avg_salary

The average salary should be rounded to 2 decimal places.

Results should be displayed in alphabetical order of department.

Input Format

TABLE SCHEMA

Table: employees

Column Name	Data Type	Description
employee_id	INT	Unique employee ID
employee_name	VARCHAR(100)	Employee Name
department	VARCHAR(50)	Department Name
salary	DECIMAL(10,2)	Employee Salary

Database Creation Script

```
CREATE TABLE employees(  
    employee_id INT PRIMARY KEY,  
    employee_name VARCHAR(100),  
    department VARCHAR(50),  
    salary DECIMAL(10,2)  
);
```

Sample Data

```
INSERT INTO employees VALUES  
(1,'Raj','IT',50000),  
(2,'Amit','IT',60000),  
(3,'Neha','HR',45000),  
(4,'Karan','HR',55000),  
(5,'Pooja','Finance',70000),  
(6,'Arun','Finance',80000);
```

Sample Input

employee_id	employee_name	department	salary
1	Raj	IT	50000
2	Amit	IT	60000
3	Neha	HR	45000
4	Karan	HR	55000
5	Pooja	Finance	70000

Output Format

Return the following columns:

Department avg_salary

Sample Output

department	avg_salary
Finance	75000.00
HR	50000.00
IT	55000.00

Explanation

Finance

$(70000 + 80000) / 2$
= 75000

HR

$(45000 + 55000) / 2$
= 50000

IT

$(50000 + 60000) / 2$
= 55000

The report displays the average salary per department.

Constraints

- Keywords, table names, column names and record values are case-sensitive.
- All records are pre-populated.
- No console input is required.
- Use exact output column names:
 - department
 - avg_salary
- Average salary must be rounded using ROUND().
- Output must be sorted by department ascending.

Final Answer

```
SELECT
    department,
    ROUND(AVG(salary),2) AS avg_salary
FROM employees
GROUP BY department
ORDER BY department;
```

TOTAL SALES BY REGION

Problem Statement

ABC Retail operates across multiple regions.

Management wants to understand how much revenue is generated from each region to support strategic planning and sales forecasting.

Task

Write an SQL query to calculate total sales amount generated in each region.

The output must contain:

- region
- total_sales

Results should be sorted alphabetically by region.

Input Format

TABLE SCHEMA

Table: sales

Column Name	Data Type	Description
sale_id	INT	Sale ID
region	VARCHAR(50)	Sales Region
amount	DECIMAL(10,2)	Sale Amount

Database Creation Script

```
CREATE TABLE sales(  
    sale_id INT PRIMARY KEY,  
    region VARCHAR(50),  
    amount DECIMAL(10,2)  
);
```

Sample Data

```
INSERT INTO sales VALUES  
(1,'North',10000),  
(2,'North',15000),  
(3,'South',12000),  
(4,'East',8000),  
(5,'West',11000),  
(6,'South',5000);
```

Sample Input

sale_id	region	amount
1	North	10000
2	North	15000
3	South	12000
4	East	8000

Output Format

Return the following columns:

Region total_sales

Sample Output

region	total_sales
East	8000
North	25000
South	17000
West	11000

Explanation

North

10000 + 15000

= 25000

South

12000 + 5000

= 17000

East

8000

West

11000

The report shows total sales generated in each region.

Constraints

- Keywords, table names, column names and record values are case-sensitive.
- All records are pre-populated.
- No console input is required.
- Use exact output column names:
 - region
 - total_sales
- Use SUM() to calculate total sales.
- Output must be sorted alphabetically by region.

Final Answer

```
SELECT
    region,
    SUM(amount) AS total_sales
FROM sales
GROUP BY region
ORDER BY region;
```

AVERAGE PRODUCT PRICE BY CATEGORY

Problem Statement

MegaMart Online Store sells products across multiple categories such as Electronics, Furniture, and Appliances.

The merchandising team wants to analyze pricing trends by calculating the average price of products in each category.

This information will help the company understand category-wise pricing and optimize product placement strategies.

Task

Write an SQL query to calculate the average product price for each category.

The output must contain:

- category
- avg_price

The average price should be rounded to 2 decimal places.

The result should be ordered alphabetically by category.

Input Format

TABLE SCHEMA

Table: products

Column Name	Data Type	Description
product_id	INT	Unique Product ID
product_name	VARCHAR(100)	Product Name
category	VARCHAR(50)	Product Category
price	DECIMAL(10,2)	Product Price

Database Creation Script

```
CREATE TABLE products(  
    product_id INT PRIMARY KEY,  
    product_name VARCHAR(100),  
    category VARCHAR(50),  
    price DECIMAL(10,2)  
);
```

Sample Data

```
INSERT INTO products VALUES  
(1,'Dell Laptop','Electronics',55000),  
(2,'HP Laptop','Electronics',60000),  
(3,'Samsung Mobile','Electronics',25000),  
(4,'Dining Table','Furniture',15000),  
(5,'Office Chair','Furniture',8000),  
(6,'Sofa','Furniture',22000);
```

Sample Input

product_id	product_name	category	price
1	Dell Laptop	Electronics	55000
2	HP Laptop	Electronics	60000
3	Samsung Mobile	Electronics	25000
4	Dining Table	Furniture	15000

Output Format

Return the following columns:

Category avg_price

Sample Output

category	avg_price
Electronics	46666.67
Furniture	15000.00

Explanation

Electronics

$(55000 + 60000 + 25000) / 3$
= 46666.67

Furniture

$(15000 + 8000 + 22000) / 3$
= 15000.00

The report displays the average product price for each category.

Constraints

- Keywords, table names, column names, and record values are case-sensitive.
 - All records are pre-populated.
 - No console input is required.
 - Use exact output column names:
 - category
 - avg_price
 - Round average values using ROUND().
 - Output must be sorted alphabetically by category.
-

Final Answer

```
SELECT
  category,
  ROUND(AVG(price),2) AS avg_price
FROM products
GROUP BY category
ORDER BY category;
```

TOTAL ORDERS BY CUSTOMER

Problem Statement

ShopEasy E-Commerce Platform wants to identify its most valuable customers. Management needs a report showing the total order value placed by each customer. This information will be used for loyalty programs and customer segmentation.

Task

Write an SQL query to calculate the total order amount for each customer.

The output must contain:

- customer_name
- total_orders

The result should be sorted by total_orders in descending order.

Input Format

TABLE SCHEMA

Table: orders

Column Name	Data Type	Description
order_id	INT	Order ID
customer_name	VARCHAR(100)	Customer Name
amount	DECIMAL(10,2)	Order Amount

Database Creation Script

```
CREATE TABLE orders(  
  order_id INT PRIMARY KEY,  
  customer_name VARCHAR(100),  
  amount DECIMAL(10,2)  
);
```

Sample Data

```
INSERT INTO orders VALUES  
(1,'Arun',1500),  
(2,'Arun',2500),  
(3,'Priya',1800),  
(4,'Rahul',3200),  
(5,'Priya',2200);
```

Sample Input

order_id	customer_name	amount
1	Arun	1500
2	Arun	2500
3	Priya	1800
4	Rahul	3200

Output Format

Return the following columns:

customer_name total_orders

Sample Output

customer_name	total_orders
Priya	4000
Arun	4000
Rahul	3200

Explanation

Arun

1500 + 2500

= 4000

Priya

1800 + 2200

= 4000

Rahul

3200

The report shows the total amount spent by each customer.

Constraints

- Keywords, table names, column names, and record values are case-sensitive.
 - All records are pre-populated.
 - No console input is required.
 - Use exact output column names:
 - customer_name
 - total_orders
 - Use SUM() for aggregation.
 - Sort by total_orders descending.
-

Final Answer

```
SELECT
    customer_name,
    SUM(amount) AS total_orders
FROM orders
GROUP BY customer_name
ORDER BY total_orders DESC;
```

MAXIMUM SALARY IN EACH DEPARTMENT

Problem Statement

TechNova Solutions wants to identify the highest-paid employee in each department.

The HR team needs a department-wise report showing the maximum salary currently being paid.

Task

Write an SQL query to find the maximum salary in each department.

The output must contain:

- department
- max_salary

TABLE SCHEMA

Table: employees

Column Name	Data Type
employee_id	INT
employee_name	VARCHAR(100)
department	VARCHAR(50)
salary	DECIMAL(10,2)

Database Creation Script

```
CREATE TABLE employees(  
    employee_id INT PRIMARY KEY,  
    employee_name VARCHAR(100),  
    department VARCHAR(50),  
    salary DECIMAL(10,2)  
);
```

Sample Data

```
INSERT INTO employees VALUES  
(1,'Raj','IT',50000),  
(2,'Amit','IT',65000),  
(3,'Neha','HR',45000),  
(4,'Karan','HR',55000),  
(5,'Pooja','Finance',80000);
```

Sample Output

department max_salary

Finance	80000
HR	55000
IT	65000

Final Answer

```
SELECT  
    department,  
    MAX(salary) AS max_salary  
FROM employees  
GROUP BY department  
ORDER BY department;
```

MINIMUM PRODUCT PRICE BY CATEGORY

Problem Statement

MegaMart wants to identify the lowest-priced product available in each category. This helps customers discover affordable options.

Task

Write an SQL query to find the minimum product price for every category.

The output must contain:

- category
 - min_price
-

TABLE SCHEMA

Table: products

| product_id | product_name | category | price |

Sample Data

INSERT INTO products VALUES

(1,'Dell Laptop','Electronics',55000),
(2,'HP Laptop','Electronics',60000),
(3,'Samsung Mobile','Electronics',25000),
(4,'Dining Table','Furniture',15000),
(5,'Office Chair','Furniture',8000);

Sample Output

category	min_price
Electronics	25000
Furniture	8000

Final Answer

```
SELECT
    category,
    MIN(price) AS min_price
FROM products
GROUP BY category
ORDER BY category;
```

COUNT EMPLOYEES BY DEPARTMENT

Problem Statement

The HR team wants to know how many employees work in each department.
This information will be used for workforce planning.

Task

Write an SQL query to count the number of employees in each department.
The output must contain:

- department
 - employee_count
-

TABLE SCHEMA

Table: employees

| employee_id | employee_name | department |

Sample Data

INSERT INTO employees VALUES

(1,'Raj','IT'),
(2,'Amit','IT'),
(3,'Neha','HR'),
(4,'Karan','HR'),
(5,'Pooja','Finance');

Sample Output

department	employee_count
Finance	1
HR	2
IT	2

Final Answer

```
SELECT
    department,
    COUNT(*) AS employee_count
FROM employees
GROUP BY department
ORDER BY department;
```

TOTAL BOOKS BY AUTHOR

Problem Statement

A publishing company wants to determine how many books each author has published. Management needs a report showing the count of books per author.

Task

Write an SQL query to count the total number of books written by each author.

The output must contain:

- author
 - total_books
-

TABLE SCHEMA

Table: books

| book_id | title | author |

Database Creation Script

```
CREATE TABLE books(  
  book_id INT PRIMARY KEY,  
  title VARCHAR(100),  
  author VARCHAR(100)  
);
```

Sample Data

```
INSERT INTO books VALUES  
(1,'Python Basics','Raj'),  
(2,'Advanced Python','Raj'),  
(3,'SQL Mastery','Priya'),  
(4,'Data Science','Priya'),  
(5,'Java Fundamentals','Amit');
```

Sample Output

author	total_books
Amit	1
Priya	2
Raj	2

Final Answer

```
SELECT  
  author,  
  COUNT(*) AS total_books  
FROM books  
GROUP BY author  
ORDER BY author;
```

AVERAGE MARKS BY CLASS

Problem Statement

A school wants to evaluate the academic performance of each class.

The principal requires a report showing the average marks obtained by students in each class.

Task

Write an SQL query to calculate the average marks of students in each class.

The output must contain:

- class_name
- avg_marks

Round the average to 2 decimal places.

TABLE SCHEMA

Table: students

| student_id | student_name | class_name | marks |

Sample Data

INSERT INTO students VALUES

(1,'Arun','10A',80),

(2,'Priya','10A',90),

(3,'Rahul','10B',75),

(4,'Neha','10B',85);

Sample Output

class_name	avg_marks
10A	85.00
10B	80.00

Final Answer

SELECT

class_name,

ROUND(AVG(marks),2) AS avg_marks

FROM students

GROUP BY class_name

ORDER BY class_name;

TOTAL REVENUE BY PRODUCT CATEGORY

Problem Statement

A retail company sells products across multiple categories.

Management wants to analyze revenue contribution from each category to identify high-performing business segments.

Task

Write an SQL query to calculate the total revenue generated by each product category.

The output must contain:

- category
 - total_revenue
-

TABLE SCHEMA

Table: sales

| sale_id | category | revenue |

Database Creation Script

```
CREATE TABLE sales(  
    sale_id INT PRIMARY KEY,  
    category VARCHAR(50),  
    revenue DECIMAL(10,2)  
);
```

Sample Data

```
INSERT INTO sales VALUES  
(1,'Electronics',50000),  
(2,'Electronics',60000),  
(3,'Furniture',30000),  
(4,'Furniture',25000),  
(5,'Clothing',15000);
```

Sample Output

category	total_revenue
Clothing	15000
Electronics	110000
Furniture	55000

Explanation

Electronics

50000 + 60000
= 110000

Furniture

30000 + 25000
= 55000

Clothing

15000

The report displays the total revenue generated by each category.

Constraints

- Keywords, table names, column names, and record values are case-sensitive.
 - All records are pre-populated.
 - No console input is required.
 - Use exact output column names:
 - category
 - total_revenue
 - Use SUM() for aggregation.
 - Output must be sorted alphabetically by category.
-

Final Answer

```
SELECT
  category,
  SUM(revenue) AS total_revenue
FROM sales
GROUP BY category
ORDER BY category;
```